

Suhaib Abdurahman

 Scholar |  LinkedIn |  Website |  Email

SUMMARY

Applied scientist with a psychology and engineering background, grounding AI systems (LLMs, agents) in psychological theory to improve performance and reliability.

SKILLS

Programming: Python, R, SQL

ML/NLP: PyTorch, Transformers/LLMs, fine-tuning, evaluation, quantized models

Agents: Tool use & structured outputs, multi-agent coordination, behavioral/persona conditioning

Statistical Analysis: Hierarchical regression, classification, clustering, time-series, Bayesian Inference

Behavioral Science: Experiments & hypothesis testing, cognition & decision-making frameworks

EXPERIENCE

Graduate Researcher, Morality & Language Lab, USC 2021–2026

- Developed multi-agent simulations to model group dynamics using persona-conditioned LLM agents.
- Ran large-scale experiments to evaluate behavioral validity and agent realism.
- Inferred latent cognitive states from agent trajectories to model decisions and downstream behaviors.
- Steered agent behavior using representations of psychological constructs (e.g., traits, values).
- Reduced simulation memory use by 60% via quantized LLMs for scalable experimentation.

LLM & GenAI Applied Science Intern, Amazon Summer 2025

- Integrated social reasoning mechanisms into AI agents, improving coordination and reliability.
- Benchmarked performance showing quantifiable improvements over strong CoT baselines.

Analyst, MicroEnergy International GmbH 2017-2019

- Delivered data-driven analyses and strategic recommendations to senior management.
- Automated financial reporting pipeline in Python, reducing manual workload and increasing accuracy.

SELECTED PUBLICATIONS

Abdurahman, Suhaib, Etsuko Ishii, et al. (2026). “Explicit Trait Inference for Multi-Agent Coordination.” In: *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics.

Abdurahman, Suhaib, Huy Vu, et al. (2024). “A deep learning approach to personality assessment: Generalizing across items and expanding the reach of survey-based research.” In: *Journal of Personality and Social Psychology* 126.2, p. 312.

Vu, Huy, **Suhaib Abdurahman**, et al. (2020). “Predicting responses to psychological questionnaires from participants’ social media posts and question text embeddings”. In: *Findings of the association for computational linguistics: Emnlp 2020*. Association for Computational Linguistics, pp. 1512–1524.

EDUCATION

PhD Social Psychology (Computational Focus), USC (2026)

M.S. Process Engineering, TU Berlin (2021)